

Statistical Quality Assurance

To impart knowledge on Quality, Quality Control And Assurance

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5123A is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5123A.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Wood and Paper Technology
Course code	5123A
Course title	Statistical Quality Assurance
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart knowledge on Quality, Quality Control And Assurance
- To introduce the concept of Statistical Quality Control.
- To illustrate the Control Chart.
- To design acceptance Sampling and O C Curve.
- Course-prerequisites:
- Knowledge of basic Mathematics Mathematics I 1
- Knowledge in Statistics Mathematics II 2

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding Assurance	See official syllabus
CO2	Analyze the Statistical quality control 14 Applying	See official syllabus
CO3	Prepare the control chart 15 Applying	See official syllabus
CO4	Solve acceptance sampling and o c curve. 14 Applying	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1
CO2	CO2 3
CO3	CO3 3

Row	Extracted official mapping
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Outline the Quality, Quality Control And Assurance	4	As specified
Module 2	Analyze the Statistical quality control	4	As specified
Module 3	Prepare the control chart	3	As specified
Module 4	Solve acceptance sampling and o c curve.	4	As specified

MODULE 1

Outline the Quality, Quality Control And Assurance

This module is presented from the official syllabus scope for Statistical Quality Assurance. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.

- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the Quality, Quality Control And Assurance
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain quality

Explain quality is an official topic in Module 1 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain quality using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain quality should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain quality means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain quality definition/meaning . steps, application . Technical terms English-

Discuss quality assurance system

Discuss quality assurance system is an official topic in Module 1 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss quality assurance system using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss quality assurance system should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss quality assurance system means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Discuss quality assurance system

താഴെ പറയുന്നവയെക്കുറിച്ച് നിങ്ങളുടെ definition/meaning എഴുതുക. നിങ്ങളുടെ നിങ്ങളുടെ, steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

Explain quality assurance program.

Explain quality assurance program. is an official topic in Module 1 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain quality assurance program. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain quality assurance program. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain quality assurance program. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain quality assurance program.

Quality assurance program definition/meaning. Quality assurance program, steps, application. Technical terms English- Malayalam.

Summarize quality in materials management 4 Understanding Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management.

Summarize quality in materials management 4 Understanding Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management. is an official topic in Module 1 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize quality in materials management 4 Understanding Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize quality in materials management 4 understanding quality assurance. quality - quality control - total quality control-quality assurance - forms of quality assurance - stages - quality planning-quality assurance programme - objectives - quality in materials management. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Summarize quality in materials management 4 Understanding Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Summarize quality in materials management 4 Understanding Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management.

definition/meaning, steps, application Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Outline the Quality, Quality Control And Assurance

M1.01	Explain quality	3
Remembering		

M1.02	Discuss quality assurance system	4
Remembering		

M1.03	Explain quality assurance program.	4
Understanding		

M1.04	Summarize quality in materials management	4
Understanding		

Contents:

Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance -

Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme

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Objectives - Quality in materials management.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Analyze the Statistical quality control

This module is presented from the official syllabus scope for Statistical Quality Assurance. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Analyze the Statistical quality control
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain statistical quality chart

Explain statistical quality chart is an official topic in Module 2 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain statistical quality chart using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain statistical quality chart should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain statistical quality chart means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain statistical quality chart definition/meaning . steps, application . Technical terms English- .

Develop frequency program.

Develop frequency program. is an official topic in Module 2 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop frequency program. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop frequency program. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop frequency program. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Develop frequency program.
 definition/meaning . , steps,
application . Technical terms English-
.

Calculate mean ,mode and median

Calculate mean ,mode and median is an official topic in Module 2 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Calculate mean ,mode and median using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, calculate mean ,mode and median should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Calculate mean ,mode and median means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- $\square$ Calculate mean ,mode and median

$\square$ $\square$ definition/meaning $\square$. $\square$ $\square$ $\square$,
steps, application $\square$ $\square$. Technical terms English- $\square$ $\square$
 $\square$.

Apply regression analysis 3 Applying Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspection and sampling Inspection.

Apply regression analysis 3 Applying Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspection and sampling Inspection. is an official topic in Module 2 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Apply regression analysis 3 Applying Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspection and sampling Inspection. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, apply regression analysis 3 applying statistical quality control. definition of sqc - characteristics - attributes and variables - process of sqc frequency distribution - frequency polygon - frequency curve - measure of central tendency - mean - median - mode - probability - normal curve - regression analysis - normal test-x2 test - comparison between 100% inspection and sampling inspection. should be discussed

with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Apply regression analysis 3 Applying Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspection and sampling Inspection. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Apply regression analysis 3 Applying Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspection and sampling Inspection. definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Analyze the Statistical quality control

M 2.01	Explain statistical quality chart Understanding	3
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M 2.02	Develop frequency program. Applying	4
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M 2.03	Calculate mean ,mode and median Understanding	4
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M 2.04	Apply regression analysis Applying	3
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	Series Test – I	1
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Contents:

Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables

- process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x² Test - Comparison between 100% inspection and sampling Inspection.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Prepare the control chart

This module is presented from the official syllabus scope for Statistical Quality Assurance. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Prepare the control chart
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain variable control chart

Explain variable control chart is an official topic in Module 3 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain variable control chart using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain variable control chart should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain variable control chart means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain variable control chart
 definition/meaning .
 application
 Technical terms English-
 .

Develop different chart in SQC

Develop different chart in SQC is an official topic in Module 3 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop different chart in SQC using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop different chart in sqc should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop different chart in SQC means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ാം Develop different chart in SQC അപ്രകാരം
അർത്ഥം definition/meaning അപ്രകാരം. അപ്രകാരം അപ്രകാരം, steps,
application അപ്രകാരം അപ്രകാരം അപ്രകാരം. Technical terms English-ൽ അപ്രകാരം
അപ്രകാരം.

Develop solution based on P chart/ U Chart 5 Applying Control Charts Control charts for Variables - X and R charts - calculation of 3σ Limits - Construction of X and r Charts - Moving Range charts - Accuracy and Precision - P chart - Pn/nP chart/U chart

Develop solution based on P chart/ U Chart 5 Applying Control Charts Control charts for Variables - X and R charts - calculation of 3σ Limits - Construction of X and r Charts - Moving Range charts - Accuracy and Precision - P chart - Pn/nP chart/U chart is an official topic in Module 3 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop solution based on P chart/ U Chart 5 Applying Control Charts Control charts for Variables - X and R charts - calculation of 3σ Limits - Construction of X and r Charts - Moving Range charts - Accuracy and Precision - P chart - Pn/nP chart/U chart using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop solution based on p chart/ u chart 5 applying control charts control charts for variables - x and r charts - calculation of 3σ limits - construction of x and r charts - moving range charts - accuracy and precision - p chart - pn/np chart/u chart should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Develop solution based on P chart/ U Chart 5 Applying Control Charts Control charts for Variables - X and R charts - calculation of 3σ Limits - Construction of X and r Charts - Moving Range charts - Accuracy and Precision - P chart - Pn/nP chart/U chart means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എന്നു വികസിപ്പിച്ചെടുക്കുന്ന പരിഹാരം P ചാർട്ട്/ U ചാർട്ട് 5 ഉപയോഗിക്കുന്ന കൺട്രോൾ ചാർട്ടുകൾ കൺട്രോൾ ചാർട്ടുകൾ വേർതിരിക്കുന്നു - X and R ചാർട്ടുകൾ - 3σ ലിമിറ്റുകൾ - X and r ചാർട്ടുകൾ - മൂവ്വിംഗ് റേഞ്ച് ചാർട്ടുകൾ - ഓർഗനൈസേഷൻ - P ചാർട്ട് - Pn/nP ചാർട്ട്/U ചാർട്ട് എന്നിവയുടെ ഔദ്യോഗിക കോഴ്സിലെ സാഹചര്യത്തിൽ നിർവ്വചനം/അർത്ഥം ഉൾക്കൊള്ളുന്നു. തത്വം, ഘടകങ്ങൾ അല്ലെങ്കിൽ ഘട്ടങ്ങൾ, ഫലം, പ്രയോഗം എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ. Technical terms English-ൽ ഉൾക്കൊള്ളുന്നു.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Prepare the control chart

M3.01	Explain variable control chart	5
Understanding		
M3.02	Develop different chart in SQC	5
Applying		
M3.03	Develop solution based on P chart/ U Chart	5
Applying		

Contents:

Control Charts Control charts for Variables - X and R charts - calculation of 3σ Limits -

Construction of X and r Charts - Moving Range charts - Accuracy and Precision - P chart -

Pn/nP chart/U chart

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Solve acceptance sampling and o c curve.

This module is presented from the official syllabus scope for Statistical Quality Assurance. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.

- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Solve acceptance sampling and o c curve.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain sampling 3 Remembering Develop the difference between sampling and

Explain sampling 3 Remembering Develop the difference between sampling and is an official topic in Module 4 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain sampling 3 Remembering Develop the difference between sampling and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain sampling 3 remembering develop the difference between sampling and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain sampling 3 Remembering Develop the difference between sampling and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain sampling 3 Remembering Develop the difference between sampling and definition/meaning. steps, application. Technical terms English-

4 Applying control chart Distinguish single sampling and double

4 Applying control chart Distinguish single sampling and double is an official topic in Module 4 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying control chart Distinguish single sampling and double using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying control chart distinguish single sampling and double should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying control chart Distinguish single sampling and double means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4- 4 Applying control chart Distinguish single sampling and double definition/meaning .
 , steps, application . Technical terms English-

3 Applying sampling plan

3 Applying sampling plan is an official topic in Module 4 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying sampling plan using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying sampling plan should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying sampling plan means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Applying sampling plan ഓരോന്നിനും ഓരോന്നിനും definition/meaning ഓരോന്നിനും. ഓരോന്നിനും ഓരോന്നിനും, steps, application ഓരോന്നിനും ഓരോന്നിനും ഓരോന്നിനും. Technical terms English-3 ഓരോന്നിനും ഓരോന്നിനും.

Apply the problems on reliability 4 Applying Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling - Limitations of sampling - Comparison between Acceptance sampling and control Charts - Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) - AOQ, Reliability - Simple Problems Text /Reference: T/R Book Title/Author T1 M.Mahajan. Published by Dhanpatharai & company R2 Industrial Engg and management by O.P. Khanna R3 M.Mahajan. Statistical Quality control ,S Chand R4 K.C.Aroa,.Total Quality Management ,Dhanapat Online Resources: SI.No Website Link 1 <https://www.youtube.com/watch?v=CfCztL5DkTE> 2 <https://www.youtube.com/watch?v=qb3mvJ1gb9g> 3 <https://www.youtube.com/watch?v=QsLDBfApqol> 4 <https://www.youtube.com/watch?v=e5g2NmIUdck> 5 https://www.youtube.com/watch?v=GC6k_qbpwmA

Apply the problems on reliability 4 Applying Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling - Limitations of sampling - Comparison between Acceptance sampling and control Charts - Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) - AOQ, Reliability - Simple Problems Text /Reference: T/R Book Title/Author T1 M.Mahajan. Published by Dhanpatharai & company R2 Industrial Engg and management by O.P. Khanna R3 M.Mahajan. Statistical Quality control ,S Chand R4 K.C.Aroa,.Total Quality Management ,Dhanapat Online Resources: SI.No Website Link 1 <https://www.youtube.com/watch?v=CfCztL5DkTE> 2 <https://www.youtube.com/watch?v=qb3mvJ1gb9g> 3 <https://www.youtube.com/watch?v=QsLDBfApqol> 4 <https://www.youtube.com/watch?v=e5g2NmIUdck> 5 https://www.youtube.com/watch?v=GC6k_qbpwmA is an official topic in Module 4 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Apply the problems on reliability 4 Applying Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling - Limitations of sampling - Comparison between Acceptance sampling and control Charts - Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) - AOQ, Reliability - Simple Problems Text /Reference: T/R Book Title/Author T1 M.Mahajan. Published by Dhanpatharai & company R2 Industrial Engg and management by O.P. Khanna R3 M.Mahajan. Statistical Quality control ,S Chand R4 K.C.Aroa,.Total Quality Management ,Dhanapat Online Resources: Sl.No Website Link 1 <https://www.youtube.com/watch?v=CfCztL5DkTE> 2 <https://www.youtube.com/watch?v=qb3mvj1gb9g> 3 <https://www.youtube.com/watch?v=QsLDBfApqol> 4 <https://www.youtube.com/watch?v=e5g2NmlUdck> 5 https://www.youtube.com/watch?v=GC6k_qbpwma using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, apply the problems on reliability 4 applying acceptance sampling and oc curve. acceptance sampling - principle of lot sampling - limitations of sampling - comparison between acceptance sampling and control charts - benefits - single sampling plan double sampling plan - oc curve - lot tolerance percent defective(ltpd) - aoq, reliability - simple problems text /reference: t/r book title/author t1 m.mahajan. published by dhanpatharai & company r2 industrial engg and management by o.p. khanna r3 m.mahajan. statistical quality control ,s chand r4 k.c.aroa,.total quality management ,dhanapat online resources: sl.no website link 1 <https://www.youtube.com/watch?v=cfcztl5dkte> 2 <https://www.youtube.com/watch?v=qb3mvj1gb9g> 3 <https://www.youtube.com/watch?v=qsldbfapqoi> 4 <https://www.youtube.com/watch?v=e5g2nmiudck> 5 https://www.youtube.com/watch?v=gc6k_qbpwma should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Apply the problems on reliability 4 Applying Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling - Limitations of sampling - Comparison between Acceptance sampling and control Charts - Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) - AOQ, Reliability - Simple Problems Text /Reference: T/R Book Title/Author T1 M.Mahajan. Published by Dhanpatharai & company R2 Industrial Engg and management by O.P. Khanna R3 M.Mahajan. Statistical Quality control ,S

Aspect	What to prepare
	Chand R4 K.C.Aroa, Total Quality Management, Dhanapat Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=CfCztL5DkTE 2 https://www.youtube.com/watch?v=qb3mvJ1gb9g 3 https://www.youtube.com/watch?v=QsLDBfApqol 4 https://www.youtube.com/watch?v=e5g2NmlUdck 5 https://www.youtube.com/watch?v=GC6k_qbpwMA means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Apply the problems on reliability 4 Applying Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling - Limitations of sampling - Comparison between Acceptance sampling and control Charts - Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) - AOQ, Reliability - Simple Problems Text /Reference: T/R Book Title/Author T1 M.Mahajan. Published by Dhanpatharai & company R2 Industrial Engg and management by O.P. Khanna R3 M.Mahajan. Statistical Quality control, S Chand R4 K.C.Aroa, Total Quality Management, Dhanapat Online Resources: Sl.No Website Link 1 <https://www.youtube.com/watch?v=CfCztL5DkTE> 2 <https://www.youtube.com/watch?v=qb3mvJ1gb9g> 3 <https://www.youtube.com/watch?v=QsLDBfApqol> 4 <https://www.youtube.com/watch?v=e5g2NmlUdck> 5 https://www.youtube.com/watch?v=GC6k_qbpwMA □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□ □□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Solve acceptance sampling and o c curve.

M4.01	Explain sampling	3
Remembering		
	Develop the difference between sampling and	
M4.02		4
Applying		
	control chart	
	Distinguish single sampling and double	
M4.03		3
Applying		
	sampling plan	
M4.04	Apply the problems on reliability	4
Applying		
	Series Test – II	1

Contents:

Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling
-
Limitations of sampling - Comparison between Acceptance sampling and control
Charts –
Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance
Percent
Defective(LTPD) - AOQ, Reliability - Simple Problems
Text /Reference:

T/R

Book Title/Author

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Explain quality. (3 marks)

Answer: Begin with the standard definition or identity of *Explain quality*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Discuss quality assurance system. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss quality assurance system*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Explain quality assurance program.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain quality assurance program..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Summarize quality in materials management 4 Understanding Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management.. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize quality in materials management 4 Understanding Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain Explain statistical quality chart. (3 marks)

Answer: Begin with the standard definition or identity of *Explain statistical quality chart*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Develop frequency program.. (3 marks)

Answer: Begin with the standard definition or identity of *Develop frequency program..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Calculate mean ,mode and median. (3 marks)

Answer: Begin with the standard definition or identity of *Calculate mean ,mode and median*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Apply regression analysis 3 Applying Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspection and sampling Inspection.. (3 marks)

Answer: Begin with the standard definition or identity of *Apply regression analysis 3 Applying Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspection and sampling Inspection..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Explain variable control chart. (3 marks)

Answer: Begin with the standard definition or identity of *Explain variable control chart.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Develop different chart in SQC. (3 marks)

Answer: Begin with the standard definition or identity of *Develop different chart in SQC.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Develop solution based on P chart/ U Chart 5 Applying Control Charts Control charts for Variables - X and R charts - calculation of 3σ Limits - Construction of X and r Charts - Moving Range charts - Accuracy and Precision - P chart - Pn/nP chart/U chart. (3 marks)

Answer: Begin with the standard definition or identity of *Develop solution based on P chart/ U Chart 5 Applying Control Charts Control charts for Variables - X and R charts - calculation of 3σ Limits - Construction of X and r Charts - Moving Range charts - Accuracy and Precision - P chart - Pn/nP chart/U chart*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain Explain sampling 3 Remembering Develop the difference between sampling and. (3 marks)

Answer: Begin with the standard definition or identity of *Explain sampling 3 Remembering Develop the difference between sampling and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Applying control chart Distinguish single sampling and double. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying control chart Distinguish single sampling and double*. State the governing principle, list the key

components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Applying sampling plan. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying sampling plan*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Apply the problems on reliability 4 Applying Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling - Limitations of sampling - Comparison between Acceptance sampling and control Charts - Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) - AOQ, Reliability - Simple Problems Text /Reference: T/R Book Title/Author T1 M.Mahajan. Published by Dhanpatharai & company R2 Industrial Engg and management by O.P. Khanna R3 M.Mahajan. Statistical Quality control ,S Chand R4 K.C.Aroa,.Total Quality Management ,Dhanapat Online Resources: Sl.No Website Link
1 <https://www.youtube.com/watch?v=CfCztL5DkTE>
2 <https://www.youtube.com/watch?v=qb3mvJ1gb9g>
3 <https://www.youtube.com/watch?v=QsLDBfApqoI>
4 <https://www.youtube.com/watch?v=e5g2NmIUdck>
5 https://www.youtube.com/watch?v=GC6k_qbpwmA. (3 marks)

Answer: Begin with the standard definition or identity of *Apply the problems on reliability 4 Applying Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling - Limitations of sampling - Comparison between Acceptance sampling and control Charts - Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) - AOQ, Reliability - Simple Problems Text /Reference: T/R Book Title/Author T1 M.Mahajan. Published by Dhanpatharai & company R2 Industrial Engg and management by O.P. Khanna R3 M.Mahajan. Statistical Quality control ,S Chand R4 K.C.Aroa,.Total Quality Management ,Dhanapat Online Resources:*

Sl.No Website Link 1 <https://www.youtube.com/watch?v=CfCztL5DkTE> 2

<https://www.youtube.com/watch?v=qb3mvJ1gb9g> 3 <https://www.youtube.com/watch?v=QsLDBfApqol> 4 <https://www.youtube.com/watch?v=e5g2NmlUdck> 5

https://www.youtube.com/watch?v=GC6k_qbpwmA. State the governing principle, list

the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain quality**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Explain statistical quality chart**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain variable control chart**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain sampling 3 Remembering Develop the difference between sampling and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.youtube.com/watch?v=CfCztL5DkTE>
<https://www.youtube.com/watch?v=qb3mvJ1gb9g> <https://www.youtube.com/watch?v=QsLDBfApqol> <https://www.youtube.com/watch?v=e5g2NmIUdck>
https://www.youtube.com/watch?v=GC6k_qbpwma

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